

CHE 201 Fall 2019 Midterm 2

Amna Nasir

TOTAL POINTS

82 / 100

QUESTION 1

1 25 pts

1.1 A 15 / 21

- 0 pts Correct
- 3 pts first is wrong
- 3 pts second is wrong
- 3 pts third is wrong
- 3 pts fourth is wrong
- 3 pts fifth is wrong
- ✓ - 3 pts sixth is wrong
- ✓ - 3 pts seventh is wrong

1.2 B 4 / 4

- ✓ - 0 pts Correct
- 2 pts explanation is not right
- 4 pts Incorrect

QUESTION 2

2 35 pts

2.1 a 7 / 10

- 0 pts Correct
- 2 pts W is not labeled
- ✓ - 2 pts System boundary is not labeled
- 1 pts Q_h is not labeled
- 1 pts Q_c is not labeled
- 4 pts Q_h and Q_c are in wrong direction
- 1 Point adjustment
- ☹ W direction is not shown

2.2 b 5 / 5

- ✓ - 0 pts Correct
- 5 pts Wrong or not done
- 2 pts calculation error

- 2 pts Q_h is wrong
- 1 pts W is not calculated

2.3 C 5 / 5

- ✓ - 0 pts Correct
- 5 pts wrong or not done
- 2 pts calculation error

2.4 d 5 / 5

- ✓ - 0 pts Correct
- 0 pts max COP is not calculated
- 2 pts calculation error, T not converted to Kelvin
- 5 pts wrong or not done
- 2 pts COP calculated but not W

2.5 e 2 / 5

- 0 pts Correct
- 5 pts wrong or not done
- ✓ - 2 pts not using heat pump is wrong or not done
- 1 pts calculation error
- 1 Point adjustment
- ☹ unit conversion is wrong

2.6 f 4 / 5

- 0 pts Correct
- 5 pts not calculated or wrong
- ✓ - 1 pts formula is wrong, should be $Q_c/T_c - Q_h/T_h = -\sigma$
- 2 pts formula is wrong
- 1 pts calculation error
- 1 pts Temperatures are not in K

QUESTION 3

3 40 pts

3.1 a 6 / 6

- ✓ - 0 pts Correct
- 3 pts turbine not shown
- 1 pts system boundary not labeled or labeled wrong
- 6 pts not done or wrong
- 1 pts turbine inlet outlet not shown

3.2 b 5 / 5

- ✓ - 0 pts Correct
- 2 pts control volume not listed
- 5 pts wrong or not done

3.3 C 4 / 4

- ✓ - 0 pts Correct
- 4 pts wrong or not done

3.4 d 7 / 8

- 0 pts Correct
- 5 pts specific volume is wrong
- ✓ - 1 pts calculation error, R unit is not right
- 8 pts not done or wrong
- 3 pts mass flow rate is wrong

3.5 e 8 / 10

- 0 pts Correct
- ✓ - 2 pts Q is not converted to kW, or sign is wrong , should be negative
- 2 pts h1 is incorrect
- 2 pts h2 is incorrect
- 10 pts not done
- 2 pts pump work not calculated
- 4 pts mass balance is wrong

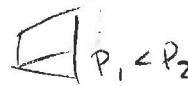
3.6 f 5 / 7

- 0 pts Correct
- 2 pts v1 is wrong or not calculated
- 1 pts v2 is wrong or not calculated
- ✓ - 2 pts calculation errors
- 7 pts not done or wrong
- 💬 this should be expansion

NAME: Amna Nasir

Please fill in answers in the space supplied and show all your work to receive full credit.
Useful Tables at the end of the Exam.

1. (21 points)

A) State whether the following statements are **True or False** (3 pts each)False A pump does work on a gas to increase the pressure and/or elevation.True Transient operation denotes a change in state with time.False As velocity decreases in a diffuser, pressure decreases. True For a closed system, entropy may be produced within the system.True The maximum coefficient of performance of any refrigeration cycle operating between cold and hot reservoirs at 40°C and 80°C, respectively, is closely 8.

$$\frac{313}{353 - 313} = 7.8 \approx 8$$

False Entropy production, σ , is path-dependent.True The second Carnot corollary states that all power cycles operating between the same two thermal reservoirs have the same thermal efficiency.

B) (4 pts) An inventor suggests an improvement on the power plant cycle. Aware that one half of the heat transferred to the steam in the furnace is discarded in the condenser to the environment, the inventor suggests getting rid of this wasteful component and sending the steam to the pump as soon as it leaves the turbine (cycle contains boiler, turbine and pump). Do you think this new design will work? Explain.

No this will not work because due to Kelvin Planck statement 100% of heat cannot be transferred to work.

All of the heated steam will not transfer 100% to work on pump.

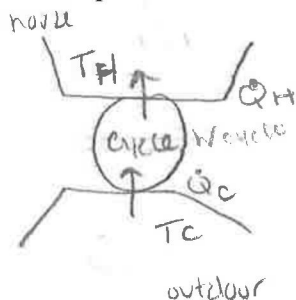
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2. (35 pts) A heat pump is used to meet the heating requirements of a house and maintain it at 20°C in winter. On a day when the average outdoor temperature drops -2°C, the house is estimated to lose heat at a rate of 80,000 kJ/h. If the heat pump under these conditions has a COP of 2.5,

- (10 pts) Draw a schematic showing the heat pump cycle, the thermal reservoirs, T_H , T_C , $\dot{W}_{cycle}$, $\dot{Q}_C$ and $\dot{Q}_H$. Show the direction of the energy transfer rates with arrows.
- (5 points) Determine the power consumed by the heat pump.
- (5 points) Determine the rate at which heat is absorbed from the cold outdoor air.
- (5 points) Determine the minimum power required to drive this heat pump.
- (5 points) Determine the total heating cost per day (12 cents/ kWh of electricity) of using heat pump. If we were to use electric resistance heater instead of heat pump, how much the cost of heating will be?
- (5 points) Determine the entropy production, σ and state if the cycle is possible, impossible or reversible. Support your answer for full credit.



$$20^\circ\text{C} = 20 + 273 = 293\text{K}$$

$$-2^\circ\text{C} = 271\text{K}$$

$$\dot{Q}_{out/H} = 80,000 \frac{\text{kJ}}{\text{h}} \times \frac{1\text{h}}{3600\text{min}} \times \frac{1\text{min}}{60\text{s}} = 22.22 \text{ kJ/s}$$

$$\text{COP}_{HP} = \frac{\dot{Q}_{out/H}}{\dot{W}_{cycle}} \Rightarrow \frac{22.22 \text{ kJ/s}}{2.5} = 2.5 \Rightarrow \frac{22.22}{2.5} = \dot{W}_{cycle} = 8.88 \text{ kJ/s} \text{ or } 8.88 \text{ kW}$$

$$\dot{W}_{cycle} = \dot{Q}_{out} - \dot{Q}_{in}$$

$$8.88 \frac{\text{kJ}}{\text{s}} = 22.22 \frac{\text{kJ}}{\text{s}} - \dot{Q}_{in} \Rightarrow \dot{Q}_{in} = 13.34 \text{ kJ/s}$$

$$\frac{T_H}{T_H - T_C} = \frac{293}{293 - 271} = 13.31 \text{ Cur max} \Rightarrow 13.31 = \frac{22.22 \text{ kJ/s}}{\dot{W}_{cycle}} \Rightarrow \dot{W}_{cycle} = 1.67 \text{ kW}$$

$$= 0.12 \times \left(8.88 \frac{\text{kJ}}{\text{s}} \times \frac{60\text{s}}{1\text{min}} \times \frac{60\text{min}}{1\text{hr}} \times \frac{24\text{hr}}{1\text{day}} \right) = \$192067.84 \text{ heat pump}$$

$$0.12 \times \left(1.67 \frac{\text{kJ}}{\text{s}} \times \frac{60}{\text{min}} \times \frac{60}{\text{hr}} \times \frac{24\text{hr}}{\text{day}} \right) = \$17314.56$$

$$-\sigma = \frac{\dot{Q}_{in}}{T_H} - \frac{\dot{Q}_{out}}{T_C} \Rightarrow -\sigma = \frac{13.34}{293} - \frac{22.22}{271} \Rightarrow -\sigma = -0.03$$

$$\sigma = 0.03 \approx 0$$

reversible

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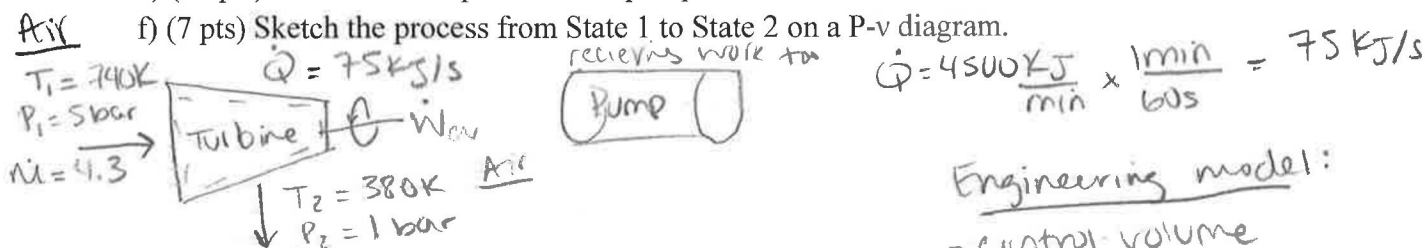
3 + W
0 -
T -
F - Q

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3. (40 points) An air turbine operating at steady state is providing power to both a pump and an air compressor. The mass flow rate of the air at the turbine inlet is 4.3 kg/s at 740K and 5 bar. The air exits the turbine at 380K and 1 bar. The turbine provides power at a rate of 850 kW to the compressor and the rate of heat transfer from the turbine to its surroundings is 4500 kJ/min. Air can be modeled as an ideal gas and kinetic and potential energy changes are negligible.

- (6 pts) Draw a schematic of the system and label the system boundary.
- (5 pts) State the engineering model
- (4 pts) Write the simplified energy rate balance
- (8 pts) Determine the volumetric flowrate of the air at the turbine exit, in m³/s.
- (10 pts) Determine the power to the pump in kW.



Engineering model:

- Control volume
- Steady state
- one inlet / one outlet
- Air is ideal gas
- $\Delta KE = \Delta PE = 0$
- work from turbine transf. to compressor.

Mass balance:

Turbine: $\frac{dm_{cv}}{dt} = \dot{m}_1 - \dot{m}_2 = 0 \Rightarrow \dot{m}_1 = \dot{m}_2 = \dot{m}$

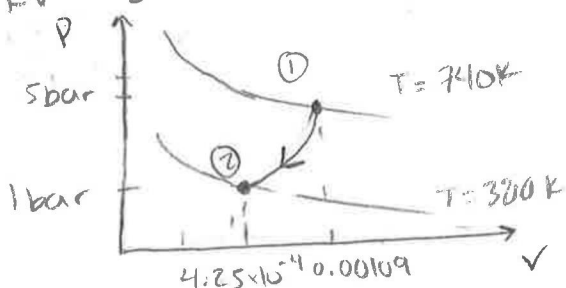
$\frac{dE_{cv}}{dt} = \dot{Q} - \dot{W} + \dot{m} [h_1 - h_2]$

$\dot{W} = \dot{Q} + \dot{m} [h_1 - h_2]$

$\dot{W} = 75 \text{ kJ/s} + 4.3 \frac{\text{kg}}{\text{s}} [756.44 - 380.77]$

$\dot{W} = 1690.381 \frac{\text{kJ}}{\text{s}}$

$1690.381 \text{ kW} - 850 \text{ kW} = 840.381 \text{ kW to Pump}$



$\dot{m} = \frac{\dot{A} \cdot \dot{V}}{v}$ (volumetric flow rate)

$Pv = RT$
 $(1 \times 10^5) v = \frac{8.314}{28.97} \cdot (380)$

$v = 0.00109$
 $4.3 \times 0.00109 = \dot{V}$
 $\dot{V}_e = 0.0047 \text{ m}^3/\text{s}$

$v_2 = 0.00109 \text{ m}^3/\text{kg}$

$Pv = RT$
 $5 \times 10^5 (v) = \frac{8.314}{28.97} (740)$

$v_1 = 4.25 \times 10^{-4} \text{ m}^3/\text{kg}$

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